# Syntecxhub_Port_Scanner

A multi-threaded TCP port scanner written in Python (standard library only).

## Features

- Scan a single host for a single port, a range, or a custom list
- Concurrent scanning with `ThreadPoolExecutor`
- Reports open / closed / timeout / error
- Logs every result to a file
- Handles bad hostnames, invalid ports, and Ctrl+C

## Usage

```
python port_scanner.py 127.0.0.1                 # ports 1-1024
python port_scanner.py 127.0.0.1 -p 20-100       # range
python port_scanner.py 127.0.0.1 -p 22,80,443    # list
python port_scanner.py scanme.nmap.org -p 1-1000 -t 200 --timeout 0.5 -v
```

| Option | Description | Default |
|---|---|---|
| `host` | Target hostname or IP | required |
| `-p, --ports` | Port, range, or list | `1-1024` |
| `-t, --threads` | Thread count | `100` |
| `--timeout` | Seconds per port | `1.0` |
| `-o, --output` | Log file | `scan_results.log` |
| `-v, --verbose` | Show closed/timeout ports | off |

## Disclaimer

For educational use only. Scan only systems you own or have explicit permission to test.